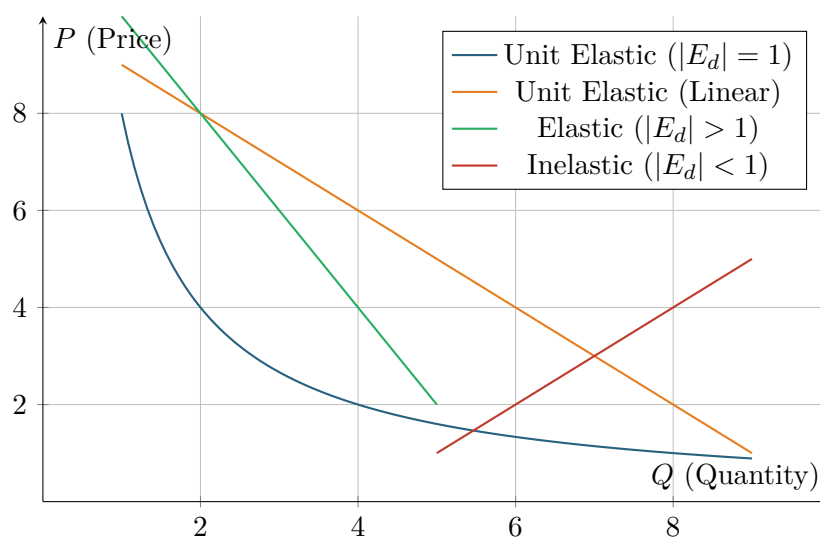


Price Elasticity of Demand

A Complete Guide with Intuition, Theory, and Practice



*“Elasticity is the economist’s way of measuring **sensitivity**.”*

30 Multiple Choice Questions Included

August 23, 2026

Contents

1 What Is Elasticity? The Core Intuition

Before we dive into formulas, let's build the intuition. Imagine you're at a coffee shop every morning, and your favorite latte costs \$5. One day, the price jumps to \$5.50. Do you still buy it? Probably—it's only 50 cents more, and you really need that caffeine. Now imagine the price jumps to \$15. You'd probably switch to tea, make coffee at home, or skip it entirely.

Elasticity measures responsiveness. It answers the question: “If I change the price, how much does the quantity demanded change?” A highly elastic good is like a sensitive scale—even a small nudge (price change) causes a big reaction (quantity change). An inelastic good is like a stubborn mule—you can push hard, and it barely moves.

Price elasticity of demand specifically measures how much the *quantity demanded* of a good responds to a change in its *price*. The formal definition:

$$\text{Price Elasticity of Demand} = \frac{\text{Percentage Change in Quantity Demanded}}{\text{Percentage Change in Price}}$$

Or in symbols:

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P}$$

Because of the law of demand (price up $\Rightarrow$ quantity down), E_d is always **negative**. However, economists typically report the **absolute value** $|E_d|$ for convenience.

1.1 Why Percentages?

You might wonder: why use percentages instead of raw units? Consider this: if the price of a car drops by \$1,000, that's huge for a \$20,000 car (5% drop) but negligible for a \$200,000 luxury car (0.5% drop). Percentages create a common language that works across all goods, big and small.

1.2 The Three Zones of Elasticity

Type	Value of $ E_d $	What It Means
Elastic	$ E_d > 1$	Quantity changes <i>more</i> than price
Unit Elastic	$ E_d = 1$	Quantity changes <i>exactly</i> as much as price
Inelastic	$ E_d < 1$	Quantity changes <i>less</i> than price

Think of it like a rubber band:

- **Elastic** (> 1): A stretchy rubber band. Pull a little (price change), and it stretches a lot (big quantity response).
- **Inelastic** (< 1): A stiff rubber band. You pull hard (big price change), and it barely stretches (small quantity response).
- **Unit Elastic** ($= 1$): A perfectly balanced rubber band. Pull it 10%, and it stretches exactly 10%.

2 The Midpoint Method

Here's a subtle but important problem. Suppose a price rises from \$10 to \$12. The percentage increase is $\frac{12-10}{10} = 20\%$. But if the price *falls* from \$12 to \$10, the percentage decrease is $\frac{10-12}{12} = -16.67\%$. The same change gives different percentages depending on direction!

Economists solve this using the **midpoint method** (also called arc elasticity). Instead of using the starting value as the base, they use the *average* of the starting and ending values.

$$E_d = \frac{\frac{Q_2 - Q_1}{(Q_2 + Q_1)/2}}{\frac{P_2 - P_1}{(P_2 + P_1)/2}}$$

In words:

- Numerator: Change in quantity divided by the *average* quantity
- Denominator: Change in price divided by the *average* price

Example: Calculating Midpoint Elasticity

A coffee shop sells 100 lattes at \$4 each. When they raise the price to \$5, sales drop to 80 lattes.

Step 1: Calculate the percentage change in quantity.

$$\frac{Q_2 - Q_1}{(Q_2 + Q_1)/2} = \frac{80 - 100}{(80 + 100)/2} = \frac{-20}{90} = -0.222 = -22.2\%$$

Step 2: Calculate the percentage change in price.

$$\frac{P_2 - P_1}{(P_2 + P_1)/2} = \frac{5 - 4}{(5 + 4)/2} = \frac{1}{4.5} = 0.222 = 22.2\%$$

Step 3: Divide.

$$E_d = \frac{-22.2\%}{22.2\%} = -1 \quad \Rightarrow \quad |E_d| = 1$$

This demand is **unit elastic**! Total revenue before: $4 \times 100 = \$400$. After: $5 \times 80 = \$400$. Revenue stayed the same, which is exactly what unit elasticity predicts.

The midpoint method is **symmetric**. Whether price goes from \$4 to \$5 or \$5 to \$4, you get the same elasticity value. This is why economists prefer it over simple percentage changes.

3 Determinants of Demand Elasticity

Why is demand for insulin inelastic while demand for restaurant meals is elastic? Several key factors determine elasticity:

3.1 Availability of Close Substitutes

This is the **single most important factor**. If there are many substitutes, consumers can easily switch when price rises.

The Substitution Principle: The more alternatives you have, the more power you have to walk away.

- **Broad category “food”:** Inelastic—you have to eat something.
- **Narrow category “ice cream”:** More elastic—you could eat cake, cookies, or frozen yogurt instead.
- **Very narrow “Ben & Jerry’s Cherry Garcia”:** Highly elastic—there are dozens of other ice cream flavors and brands.

3.2 Luxuries vs. Necessities

- **Necessities** (insulin, basic food, electricity): Inelastic. You need them regardless of price.
- **Luxuries** (yachts, first-class flights, designer handbags): Elastic. When prices rise, you simply do without.

Ask yourself: “Could I survive without this?” If the answer is no, demand is probably inelastic. If yes, it’s probably elastic.

3.3 Market Definition

The more narrowly a market is defined, the more elastic demand becomes.

Market Definition	Typical Elasticity
All beverages	Very inelastic
Soft drinks	Inelastic
Cola drinks	Somewhat elastic
Coca-Cola	Elastic
Diet Cherry Vanilla Coca-Cola	Very elastic

3.4 Time Horizon

Demand becomes **more elastic over time**. In the short run, habits and existing contracts lock people in. In the long run, people find alternatives, change behavior, and develop new habits.

Gasoline Example:

- **This week:** Gas prices jump 20%. You still drive to work. Demand is highly inelastic.
- **This year:** You start carpooling, take the bus, and plan trips better. Demand becomes more elastic.
- **Five years:** You buy a hybrid car, move closer to work, or work remotely. Demand is now quite elastic.

3.5 Share of Income

Goods that consume a large portion of your budget tend to have more elastic demand. A 10% increase in rent matters far more than a 10% increase in salt, even though both are necessities.

4 Total Revenue and Elasticity

Total revenue (TR) is simply:

$$TR = P \times Q$$

This is the total amount paid by buyers and received by sellers. The fascinating thing about elasticity is that it tells you **exactly what happens to total revenue when price changes**.

4.1 The Revenue-Elasticity Relationship

Demand Type	Elasticity	Price $\uparrow \Rightarrow$ TR ...
Inelastic	$ E_d < 1$	Increases (price and TR move together)
Unit Elastic	$ E_d = 1$	Stays the same
Elastic	$ E_d > 1$	Decreases (price and TR move opposite)

The Elasticity-Revenue Story:

When you raise price, two things happen:

1. **Price effect:** You get more money per unit sold (good for revenue).
2. **Quantity effect:** You sell fewer units (bad for revenue).

Which effect wins? Elasticity decides:

- **Inelastic:** The quantity effect is tiny. Few people leave, so the price effect dominates. Revenue rises.
- **Elastic:** The quantity effect is huge. Many people leave, overwhelming the price gain. Revenue falls.
- **Unit Elastic:** The two effects perfectly cancel. Revenue is unchanged.

Pharmaceutical Example:

A drug company sells 10,000 pills at \$10 each ($TR = \$100,000$).

Case A: Inelastic demand ($|E_d| = 0.5$)

Price rises to \$12 (20% increase). Quantity falls by only 10% (to 9,000).

$$TR = 12 \times 9,000 = \$108,000 \quad (\text{Revenue went UP!})$$

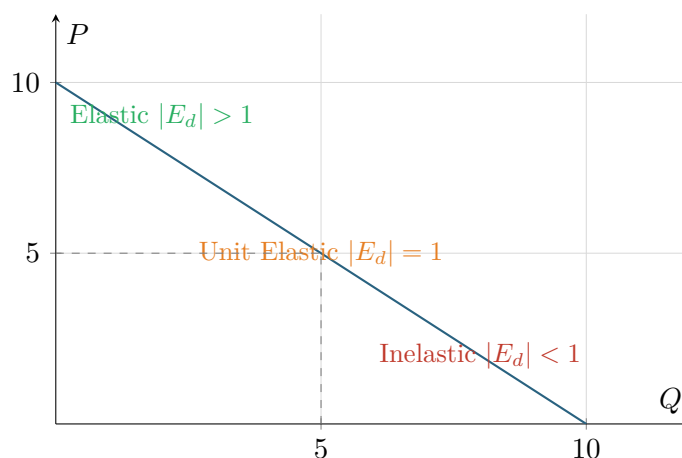
Case B: Elastic demand ($|E_d| = 2$)

Price rises to \$12 (20% increase). Quantity falls by 40% (to 6,000).

$$TR = 12 \times 6,000 = \$72,000 \quad (\text{Revenue went DOWN!})$$

4.2 The Linear Demand Curve Paradox

On a straight-line demand curve, elasticity is **not constant**! It varies from elastic at the top to inelastic at the bottom.



At the top of the demand curve, prices are high and quantities are low. A price change represents a small percentage change (because the base price is high), while the quantity change is a large percentage (because the base quantity is low). This makes demand elastic at high prices.

At the bottom, the opposite is true: price changes are large percentages, quantity changes are small percentages. Demand becomes inelastic.

5 Other Elasticity Concepts

5.1 Income Elasticity of Demand

This measures how quantity demanded responds to changes in **consumer income**.

$$E_I = \frac{\% \Delta Q_d}{\% \Delta I} = \frac{\frac{Q_2 - Q_1}{(Q_2 + Q_1)/2}}{\frac{I_2 - I_1}{(I_2 + I_1)/2}}$$

Type	E_I Value	Examples
Normal Good	$E_I > 0$	Restaurant meals, cars, vacations
Luxury (Normal)	$E_I > 1$	Yachts, caviar, designer clothes
Necessity (Normal)	$0 < E_I < 1$	Basic food, utilities
Inferior Good	$E_I < 0$	Ramen noodles, bus rides, used clothes

Intuition

Normal goods: As you get richer, you buy more. Most goods are normal.

Inferior goods: As you get richer, you buy *less*. Not because they're bad, but because

you can afford better alternatives. When your income rises, you might switch from instant ramen to fresh pasta, or from the bus to Uber.

5.2 Cross-Price Elasticity of Demand

This measures how the quantity demanded of Good A responds to a price change in Good B.

$$E_{XY} = \frac{\% \Delta Q_X}{\% \Delta P_Y}$$

Value	Relationship	Example
$E_{XY} > 0$	Substitutes	Coffee and tea
$E_{XY} < 0$	Complements	Coffee and sugar
$E_{XY} = 0$	Unrelated	Coffee and shoes

Substitutes ($E_{XY} > 0$): If coffee gets expensive, people buy more tea. The price of one goes up, demand for the other goes up. Positive relationship.

Complements ($E_{XY} < 0$): If coffee gets expensive, people buy less sugar (because they're making less coffee). The price of one goes up, demand for the other goes down. Negative relationship.

5.3 Price Elasticity of Supply

This measures how **quantity supplied** responds to price changes.

$$E_s = \frac{\% \Delta Q_s}{\% \Delta P}$$

Unlike demand elasticity, supply elasticity is **always positive** (higher prices incentivize more production).

Time is the key determinant of supply elasticity:

- **Short run:** Supply is often inelastic. Factories can't be built overnight. Farmers can't grow more crops instantly.
- **Long run:** Supply becomes much more elastic. New firms enter, existing firms expand, technology improves, and resources can be reallocated.

6 Real-World Applications

6.1 Farming Technology and the Farmer's Paradox

Advances in agricultural technology (better seeds, machinery, irrigation) increase crop supply. This should be good for farmers, right? Not necessarily.

The Paradox: Food demand is highly **inelastic** (people only eat so much). When supply increases, prices drop *sharply*. Because demand is inelastic, the price drop causes total revenue for farmers to *fall*.

Example: A new fertilizer doubles wheat production. But since everyone was already eating enough bread, the extra wheat just crashes the price. Farmers end up worse off collectively.

This is why governments often implement **price supports** and **supply restrictions** (like paying farmers *not* to grow crops)—to prevent this revenue collapse.

6.2 OPEC and Oil Prices

OPEC (Organization of the Petroleum Exporting Countries) successfully raised oil prices in the 1970s by restricting supply.

Why it worked (short run): Both oil supply and demand were highly inelastic in the short run. Consumers couldn't quickly buy electric cars or insulate homes. Producers outside OPEC couldn't ramp up production instantly. Restricting supply caused prices to soar with minimal quantity reduction.

Why it failed (long run): Over time, consumers bought fuel-efficient cars, switched to natural gas, and developed alternative energy. Non-OPEC producers expanded drilling. Both supply and demand became more elastic. OPEC's price-fixing power diminished because the market adapted.

6.3 Drug Interdiction Policy

Governments try to reduce drug use by interdicting (seizing) supplies, which reduces supply and raises prices.

The Unintended Consequence: Drug demand is highly inelastic, especially in the short run (addiction means users will pay almost any price). When interdiction raises prices:

1. Total spending on drugs **increases** (because $P \uparrow$ and Q barely falls).
2. Addicts need more money to buy drugs.
3. Drug-related crime (theft, robbery) **increases** to fund the higher spending.

This is a classic example of how ignoring elasticity can lead to policies with the opposite of the intended effect. Some economists argue that **treatment and education** (reducing demand) would be more effective than **interdiction** (reducing supply) for inelastic goods.

7 Summary Table

Concept	Formula	Key Intuition
Price Elasticity of Demand	$\frac{\% \Delta Q_d}{\% \Delta P}$	How sensitive buyers are to price changes
Income Elasticity	$\frac{\% \Delta Q_d}{\% \Delta I}$	How demand changes when people get richer/poorer
Cross-Price Elasticity	$\frac{\% \Delta Q_x}{\% \Delta P_y}$	Whether goods are substitutes (+) or complements (−)
Price Elasticity of Supply	$\frac{\% \Delta Q_s}{\% \Delta P}$	How responsive producers are to price changes
Total Revenue Rule	$TR = P \times Q$	For inelastic goods, raise prices; for elastic goods, lower prices

8 Practice Problems: 30 Multiple Choice Questions

Instructions: Choose the best answer for each question. Answers are provided at the end.

1. If the price elasticity of demand for a good is -2.5 , the demand is:
 - (a) perfectly elastic
 - (b) elastic
 - (c) unit elastic
 - (d) inelastic
 - (e) perfectly inelastic

2. The midpoint method for calculating elasticity is preferred because it:
 - (a) uses calculus instead of algebra
 - (b) gives the same result regardless of the direction of change
 - (c) is required by law in economic analysis
 - (d) only works for linear demand curves
 - (e) assumes demand is always unit elastic

3. If demand is inelastic and a firm raises its price, total revenue will:
 - (a) decrease
 - (b) increase
 - (c) remain unchanged
 - (d) become zero
 - (e) first increase then decrease

4. Which of the following would make demand for a good *more elastic*?
 - (a) The good is a necessity
 - (b) There are no close substitutes
 - (c) The market is broadly defined
 - (d) Consumers have more time to adjust
 - (e) The good represents a small share of income

5. A linear demand curve has elasticity that:
 - (a) is constant throughout
 - (b) is elastic at high prices and inelastic at low prices

- (c) is inelastic at high prices and elastic at low prices
- (d) equals zero everywhere
- (e) equals infinity everywhere

6. The price of apples rises from \$2 to \$3, and quantity demanded falls from 100 to 80. Using the midpoint method, the price elasticity of demand is approximately:

- (a) -0.56
- (b) -0.44
- (c) -1.0
- (d) -2.25
- (e) -1.5

7. If the cross-price elasticity between goods X and Y is $+0.8$, then X and Y are:

- (a) complements
- (b) substitutes
- (c) unrelated goods
- (d) normal goods
- (e) inferior goods

8. An inferior good has an income elasticity that is:

- (a) positive and greater than 1
- (b) positive and less than 1
- (c) exactly zero
- (d) negative
- (e) exactly 1

9. If demand is unit elastic and price increases by 10%, quantity demanded will:

- (a) increase by 10%
- (b) decrease by 10%
- (c) increase by more than 10%
- (d) decrease by less than 10%
- (e) remain unchanged

10. Which good is likely to have the most inelastic demand?

- (a) A specific brand of soda

- (b) Soda in general
- (c) Beverages in general
- (d) Insulin for diabetics
- (e) Restaurant meals

11. Total revenue is maximized when demand is:

- (a) perfectly elastic
- (b) elastic
- (c) unit elastic
- (d) inelastic
- (e) perfectly inelastic

12. If the income elasticity of demand for a good is -0.5 , the good is:

- (a) a normal necessity
- (b) a normal luxury
- (c) an inferior good
- (d) a substitute good
- (e) a complement good

13. The price elasticity of supply tends to be *more elastic* in the long run because:

- (a) consumers have more time to find substitutes
- (b) producers can build new factories and enter the market
- (c) the government regulates long-run prices
- (d) demand curves shift left over time
- (e) supply curves always become horizontal

14. A farmer's new technology doubles wheat production. If demand for wheat is inelastic, the farmer's total revenue will:

- (a) double
- (b) increase but less than double
- (c) remain the same
- (d) decrease
- (e) become zero

15. If two goods are complements, their cross-price elasticity of demand is:

- (a) positive
- (b) negative
- (c) zero
- (d) greater than 1
- (e) equal to 1

16. A horizontal demand curve has a price elasticity of demand that is:

- (a) zero
- (b) between 0 and 1
- (c) exactly 1
- (d) greater than 1
- (e) infinite

17. If a 5% increase in price leads to a 2% decrease in quantity demanded, demand is:

- (a) elastic
- (b) unit elastic
- (c) inelastic
- (d) perfectly elastic
- (e) perfectly inelastic

18. OPEC was able to raise oil prices significantly in the 1970s primarily because:

- (a) oil supply was perfectly elastic
- (b) both oil supply and demand were inelastic in the short run
- (c) consumers immediately switched to electric cars
- (d) new oil producers entered the market quickly
- (e) the government subsidized oil production

19. The “paradox of plenty” in agriculture occurs because:

- (a) farmers don’t understand economics
- (b) increased supply with inelastic demand lowers total revenue
- (c) technology always destroys jobs
- (d) consumers eat less when food is cheaper
- (e) the government taxes agricultural output

20. If demand is elastic and a firm wants to increase total revenue, it should:

- (a) raise price
- (b) lower price
- (c) keep price constant
- (d) advertise more
- (e) reduce quality

21. A vertical demand curve represents demand that is:

- (a) perfectly elastic
- (b) unit elastic
- (c) perfectly inelastic
- (d) elastic
- (e) income elastic

22. The price elasticity of demand for a narrowly defined good (e.g., “Coca-Cola”) is typically *more elastic* than for a broadly defined good (e.g., “beverages”) because:

- (a) broadly defined goods have higher prices
- (b) narrowly defined goods have more substitutes
- (c) broadly defined goods are necessities
- (d) narrowly defined goods are luxuries
- (e) broadly defined goods have lower quality

23. If the price of good X increases from \$10 to \$12 and quantity demanded decreases from 200 to 150, total revenue:

- (a) increases from \$2,000 to \$2,400
- (b) decreases from \$2,000 to \$1,800
- (c) remains at \$2,000
- (d) increases from \$2,000 to \$1,800
- (e) decreases from \$2,400 to \$2,000

24. Drug interdiction policies that reduce drug supply may increase drug-related crime because:

- (a) drug demand is elastic, so users buy more
- (b) drug demand is inelastic, so total spending on drugs rises
- (c) drug dealers become more violent

- (d) the police are corrupt
- (e) drug prices fall, attracting new users

25. If income rises by 10% and the quantity demanded of a good rises by 15%, the good is:

- (a) an inferior good
- (b) a normal necessity
- (c) a normal luxury
- (d) a complement
- (e) a substitute

26. On a linear demand curve, total revenue is maximized at the point where:

- (a) price is highest
- (b) quantity is highest
- (c) demand is unit elastic
- (d) demand is perfectly elastic
- (e) demand is perfectly inelastic

27. The price elasticity of supply measures:

- (a) how quantity demanded responds to income changes
- (b) how quantity supplied responds to price changes
- (c) how price responds to quantity supplied changes
- (d) how quantity demanded responds to price changes
- (e) how supply responds to demand changes

28. If the cross-price elasticity between peanut butter and jelly is -0.4 , a 10% increase in the price of peanut butter will cause the quantity demanded of jelly to:

- (a) increase by 4%
- (b) decrease by 4%
- (c) increase by 25%
- (d) decrease by 25%
- (e) remain unchanged

29. A good with an income elasticity of $+0.3$ is classified as:

- (a) an inferior good
- (b) a normal luxury

- (c) a normal necessity
- (d) a perfect substitute
- (e) a perfect complement

30. Which statement about elasticity is **false**?

- (a) Elastic demand means $|E_d| > 1$
- (b) The midpoint method ensures symmetric elasticity calculations
- (c) Demand for luxuries tends to be more elastic than demand for necessities
- (d) Supply is always more elastic than demand
- (e) Time horizon affects both demand and supply elasticity

9 Answer Key

Q	Answer	Q	Answer	Q	Answer
1	(b)	11	(c)	21	(c)
2	(b)	12	(c)	22	(b)
3	(b)	13	(b)	23	(b)
4	(d)	14	(d)	24	(b)
5	(b)	15	(b)	25	(c)
6	(a)	16	(e)	26	(c)
7	(b)	17	(c)	27	(b)
8	(d)	18	(b)	28	(b)
9	(b)	19	(b)	29	(c)
10	(d)	20	(b)	30	(d)

Detailed Explanations for Selected Questions

Q1: $|E_d| = 2.5 > 1$, so demand is **elastic**. The negative sign just reflects the law of demand.

Q6: $\frac{(80-100)/90}{(3-2)/2.5} = \frac{-20/90}{1/2.5} = \frac{-0.222}{0.4} \approx -0.56$.

Q16: A horizontal demand curve means consumers will buy any quantity at a fixed price but zero at any higher price. This is **perfectly elastic** ($E_d = \infty$).

Q23: $TR_{\text{before}} = 10 \times 200 = \$2,000$. $TR_{\text{after}} = 12 \times 150 = \$1,800$. Revenue decreased, which is consistent with elastic demand.

Q28: Cross-price elasticity is negative (-0.4), so these are complements. A price increase in peanut butter reduces jelly demand: $-0.4 \times 10\% = -4\%$.

Q30: Supply is **not** always more elastic than demand. It depends on the good, time horizon, and market conditions. This is the false statement.

“The most important single central fact about a free market is that no exchange takes place unless both parties benefit.”

— Milton Friedman